Records & patterns · my notes [16

a value is **exactly one of** a known set

Sealed = a closed box of shapes 📦

sealed PaymentResult permits ...

Approved	Declined	Pending	Refunded
id, method, amount	id, method, reason	id, since	id, amount
8 payments	2 payments	P008	P006

each box carries only ITS data · no nullable "sometimes" fields · add a 5th box → compiler lists every switch to fix

Taking it apart 🔧

`Approved("P005", "PAYNOW", 1500.00)` →

```
case Approved(var id, var m, var amt)
when amt ≥ 1000 → "P005 APPROVED large"
```

general "case Approved a" first? → compile error (dominated)

Records 🧾

compact constructor: validate + scale 2

`Money.of("SGD", "10.005")` → SGD 10.01

SGD + USD → currency mismatch

local record inside a method [16]

Evolution 🪜

8: instanceof + cast

14: switch expressions

16: instanceof patterns, records

17: sealed · 21: pattern switch

case null [21] 🕳

`case null -> "null!"`

no NullPointerException from switch

over Object: default required

over sealed: no default

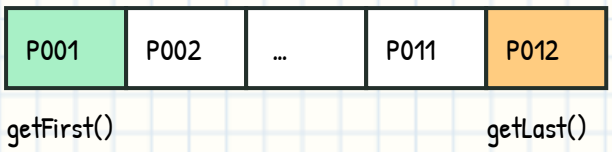
Streams + switch 🌊

`groupingBy(r -> switch (r) {..})`

{APPROVED=8, DECLINED=2, PENDING=1, REFUNDED=1}

settled (Approved only) = 2999.19

First, last, reversed [21] ↔



reversed() → [P012, P011, ...]

a VIEW: changes show through

maps: firstEntry · pollFirstEntry

Worked example: payment outcome report

BEFORE: if/else on status + final else catch-all · AFTER [8]: abstract Outcome + 4 subclasses (open) · MODERN: sealed records + exhaustive switches · 12 lines + APPROVED=8 DECLINED=2 PENDING=1 REFUNDED=1 settled=2999.19 · all 3 agree ✓

Exercise: claims 🏥

Approved · PartiallyApproved · Rejected · UnderReview

CL-03: 12000 → PARTIAL 9600.00

total paid 35280.50

Beyond Java 21 🔭

unnamed patterns `_` : Java 22

primitive patterns: preview

this course: Java 21 only

Claim rules (course rules) 📋

unknown policy → Rejected · incident after expiry → Rejected · documents missing → UnderReview · claimed > 10 × premium → PartiallyApproved(10 × premium) · else Approved(claimed) · {APPROVED=3, PARTIAL=2, REJECTED=2, UNDER_REVIEW=1}

★ Remember

validate in compact constructors · one record per alternative · seal the set · no default · guarded cases first · case null · getFirst/getLast/reversed · next → L17 functional error handling (Result = Success | Failure)